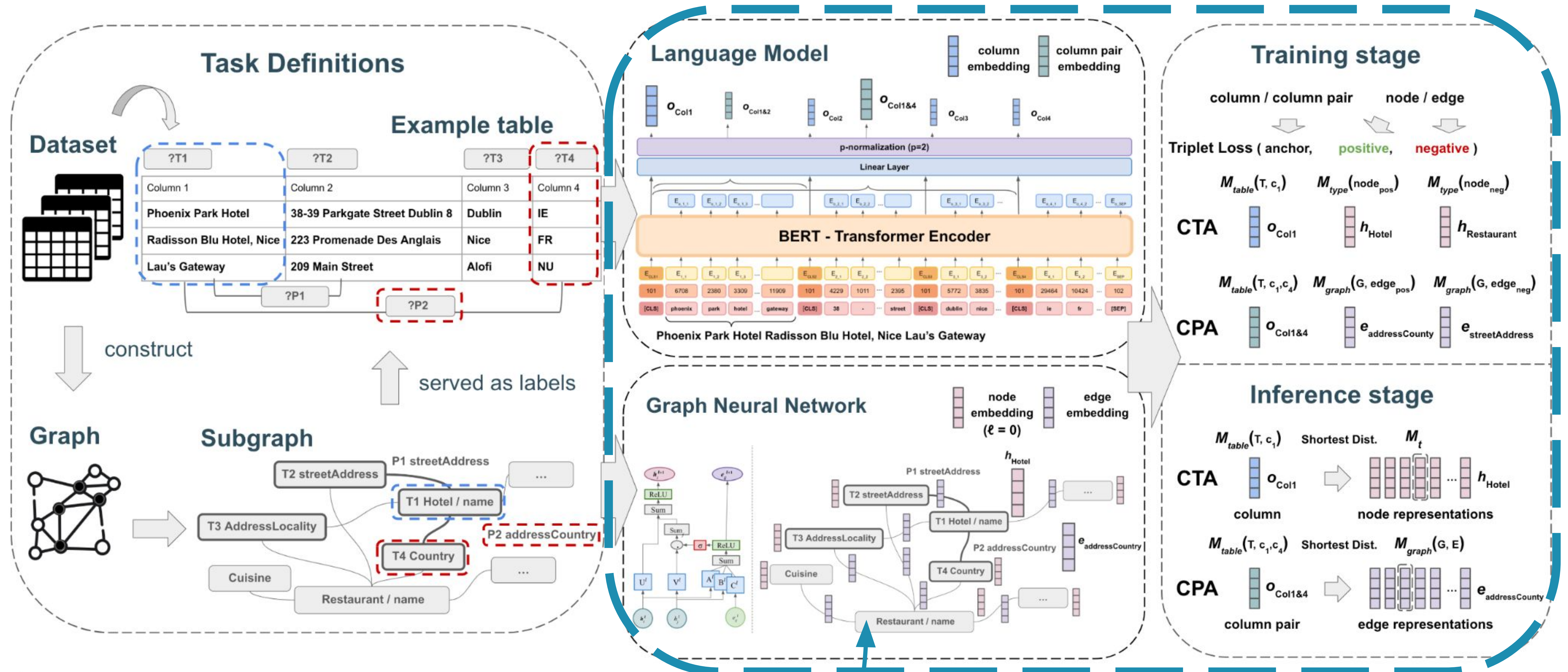




Label-Constrained Column Annotation with Language Models and Graph Neural Networks



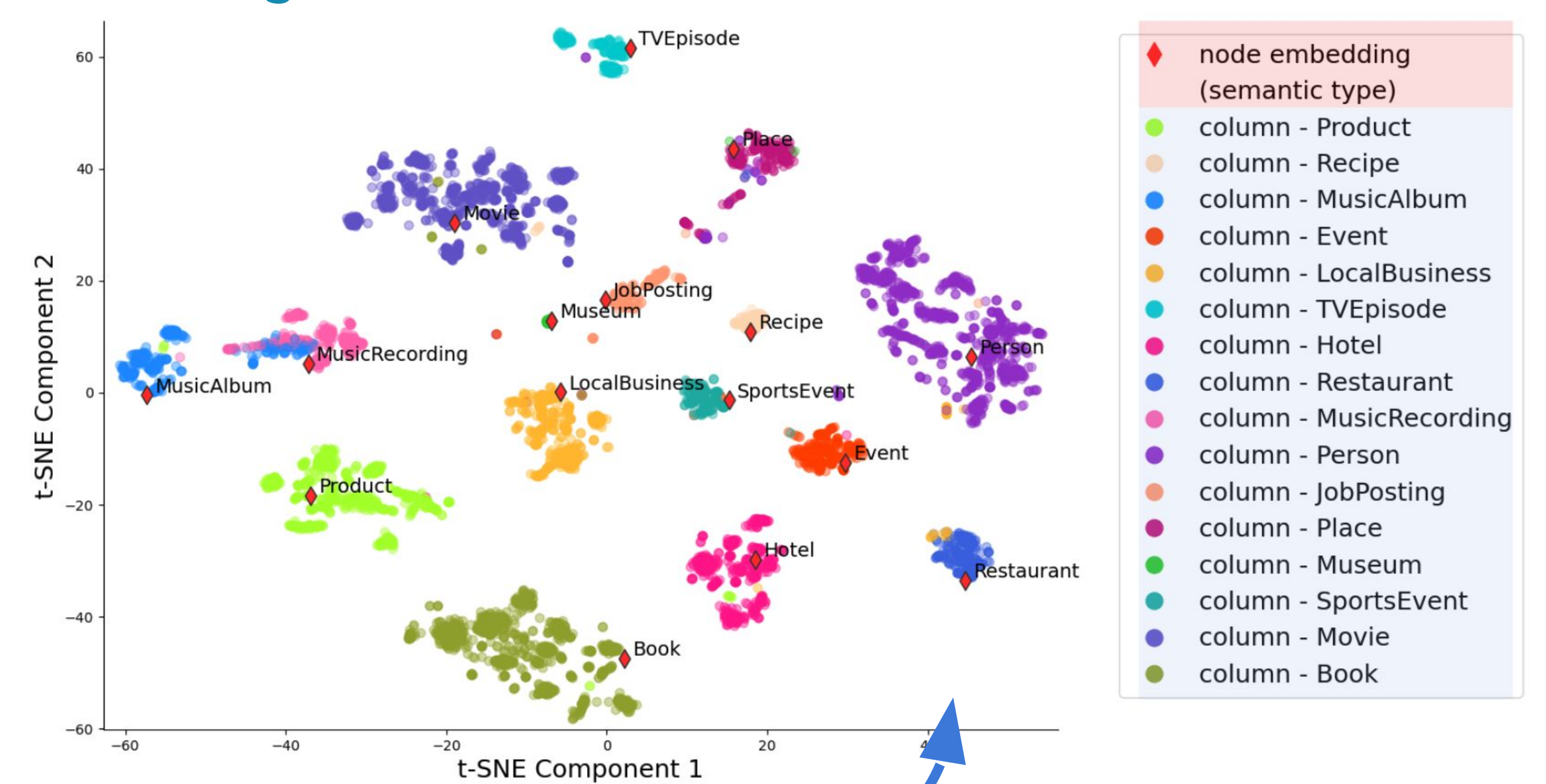
Motivations

Previous works treat table understanding as a classification problem, using language models trained on annotated tables with type and property labels, but **they overlook the graph structure of the label space**.

We propose **RODEO**, a **two-tower** embedding-based training framework:

- matching **columns / column pair** in the table to **nodes / edges** in the graph.
- **language models** can effectively model the content **within individual tables**.
- **GNN** can encode labels (constraints) **across individual tables**.

Two-tower training framework



The training process brings semantically related columns and labels closer in the embedding space



Methodology

In RODEO, **columns, column pairs, and semantic labels are all represented as embeddings, requiring training with triplet loss**.

- column / column pair serves as **anchor**, the ground truth serves as **positive**.
- an **online negative mining strategy** is applied to selects the **negative**.
- an **ablation study compares triplet loss with cross-entropy loss** (without GNN)



Performance & findings

- **Triplet loss**, combined with our negative mining strategy, is **self-standing and effective for learning embeddings** in semantic table annotation.
- RODEO outperformed the state-of-the-art methods, while **adding graph constraints** consistently leads to **better performance** in the tasks.
- RODEO can handle many **real-world** tabular challenges, e.g., **annotating tables with a massive number of numerical columns**.
- RODEO, as a two-tower framework that is **agnostic to the table encoder** supports **scaling from small language models like BERT to large language models (LLMs)** that are capable of generating embeddings.

Method	CTA		CPA	
	Macro F1	Micro F1	Macro F1	Micro F1
DODUO	67.15	71.77	66.10	72.94
w/o GNN	66.84	72.25	66.68	74.3
w/ GNN ($\ell = 1$)	71.32	73.44	66.88	73.34
w/ GNN ($\ell = 2$)	72.72	75.16	65.40	72.71
w/ GNN ($\ell = 3$)	73.55	76.04	69.79	75.08
w/ GNN ($\ell = 4$)	72.59	74.70	66.01	73.78
w/o synthetic properties	73.24	75.03	68.45	75.44
w/ all synthetic properties	71.74	74.38	67.52	73.66

Scaling up RODEO from BERT to LLMs, model variant using Qwen3-8B embeddings outperformed all other zero-shot and fine-tuned GPT models in relation annotation

Method	Params	Setting	Micro F1
TURL	14.5M	fine-tuned	60.75
DODUO	110M	fine-tuned	70.34
RoBERTA	110M	fine-tuned	71.45
GPT-3.5	↑ (B)	zero-shot	66.49
GPT-3.5	↑ (B)	fine-tuned	83.55
GPT-4	↑↑ (B)	zero-shot	81.43
RODEO - BERT	~110M	fine-tuned	81.54
RODEO - Qwen3	~0.6B	fine-tuned	83.15
RODEO - Qwen3	~4B	fine-tuned	83.76
RODEO - Qwen3	~8B	fine-tuned	85.09

* Closed-source LLMs do not disclose parameters [56], estimated at billion scale.